SANMUGAPPRIYAN S

Entry-Level RTL Design / Digital Design Engineer | Verilog HDL | VHDL

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GitHub: github.com/ssanmugappriyan/Sanmugappriyan

PROFILE SUMMARY

Entry-level RTL Design / Digital Design Engineer candidate (B.E. Electronics and Communication Engineering, graduating 2026) targeting semiconductor and VLSI opportunities. Designed an 8-bit, 16-entry synchronous FIFO in Verilog HDL with read/write pointer management, occupancy counter and full/empty flag logic, simulated with Icarus Verilog and GTKWave; testbench coverage is currently being expanded. Strong foundation in digital logic design, combinational and sequential circuits and simulation-based verification, with VHDL knowledge. Completed a hardware R&D internship in validation, simulation output analysis and signal-level debugging.

TECHNICAL SKILLS

- **HDL:** Verilog HDL, VHDL
- **Digital Design:** Digital logic design, combinational and sequential circuits, RTL design fundamentals, FIFO architecture, memory and pointer management
- **Verification:** Testbench development, simulation-based functional verification, waveform inspection
- **Simulation Tools:** Icarus Verilog, GTKWave, ModelSim, OrCAD PSpice, OrCAD Capture CIS
- **Programming:** C, Embedded C (STM32F446 HAL, ESP32-C3), Python (basic)

PROJECTS

Synchronous FIFO Design Using Verilog HDL | Icarus Verilog, GTKWave | Design implemented; verification in progress | github.com/ssanmugappriyan/Sanmugappriyan (FIFO folder)

- Designed an 8-bit, 16-entry synchronous FIFO in Verilog HDL, implementing the memory as a register array with read and write pointer management and occupancy counter logic.
- Implemented full and empty status flag generation and a synchronous active-high reset.
- Developed a Verilog testbench and simulated the design with Icarus Verilog, inspecting waveforms in GTKWave.
- Verified basic single write/read operations and FIFO capacity through simulation.
- In progress: expanding the testbench to cover multiple transactions and additional corner cases.

Smart Knee Rehabilitation Monitoring Device | ESP32-C3, MPU6050, Embedded C, I2C | College Project

- Developed Embedded C firmware for a wearable device measuring real-time knee flexion and extension angles.
- Interfaced the MPU6050 IMU over I2C, including register-level initialization, and debugged the hardware using I2C bus analysis and power verification.

WORK EXPERIENCE

Hardware R&D Intern | Qmax Test Equipments Pvt Ltd | June – September 2025

- Executed structured validation and verification test cases on electronic hardware modules and analyzed simulation outputs against expected design behavior.
- Analyzed signal behavior and test results to identify and debug functional issues.
- Developed STM32F446 firmware (STM32 HAL) for a working three-tank level monitoring prototype with GPIO input handling and LED status indication.
- Redesigned an existing STM32 hardware module by removing unnecessary components, then assembled a simplified prototype and evaluated its functionality.
- Interpreted circuit schematics and datasheets and documented verification results and design changes.

EDUCATION

B.E. Electronics and Communication Engineering | Saveetha School of Engineering | 2022 – 2026

CGPA: 8.0 / 10

Senior Secondary | Sri Chaitanya Techno School | 2021 – 2022

CERTIFICATION

- OrCAD X Capture v24.1: official digital badge validating schematic design skills